Question 1: *Proof.* Suppose $a, b, c \in \mathbb{Z}^+$ and that $a \mid c$ and $b \mid c$. Suppose also that $m = lcm(a, b)$. By Theorem 5.23 in the notes, there must exist integers $q, r \in \mathbb{Z}$ with $0 \leq r < m$ such that $c = qm + r$. $a \mid m$ and $b \mid m$, and so combined with our assumptions about c, $a \mid r$ and $b \mid r$, so r is a multiple of a and b. However, $r < m$, so $r = 0$. $r = 0 \implies m \mid c$. □

Question 2: *Proof.* Let $p, q \in \mathbb{Z}^+$ be prime, and $p \mid q$. $p \mid q$ means that $q = kp, k \in \mathbb{Z}$. $q \mid kp$, and by the definition of primality, we must have $q \mid p$ or $q \mid k$. If $q \mid p$, we are done because now we have $p \mid q$ and $q \mid p$, and by the antisymmetry of $\mid$, $p = q$. If $q \mid k$, then, $k = dq, d \in \mathbb{Z}$. Plugging in to $q = kp$, get get $q = (dq)p$. Cancelling, we get $1 = dp$. Over the integers, this only occurs when $p = \pm 1$, which is a contradiction because neither -1 or 1 is a prime, and we have p prime by assumption. Thus, we are done. □

Question 3: Rearranging, we have $lcm(a, b) = \frac{ab}{\gcd(a,b)}$. By the Euclidean algorithm, we get that $\gcd(1495, 650) = 65$. Plugging in, we get that $lcm(a, b) = \frac{1495 \cdot 650}{65} = 14,950$.

Using prime factorization, we have that the prime factorization of 1495 and 650 is $1495 = 5 \cdot 13 \cdot 23$ and $650 = 2 \cdot 5^2 \cdot 13$. Using the least common multiple algorithm, we take term of every prime factor that appears in either 1495 or 650 that has the maximum power between the two. Thus, we have $2 \cdot 5^2 \cdot 13 \cdot 23 = 14,950$ also.

Question 4: *Proof.* By the Fundamental Theorem of Arithemtic, $b^2 = {p_1}^{2f_1} \cdots {p_k}^{2f_k}$ and $a^2 = {p_1}^{2e_1} \cdots {p_k}^{2e_k}$. Also, $b^2 = na^2, n \in \mathbb{Z}$. We can also write n as a product of primes, so we get

$${p_1}^{2f_1} \cdots {p_k}^{2f_k} = ({p_1}^{d_1} \cdots {p_k}^{d_k})({p_1}^{2e_1} \cdots {p_k}^{2e_k}).$$

Expanding, we get that

$${p_1}^{2f_1} \cdots {p_k}^{2f_k} = {p_1}^{2e_1+d_1} \cdots {p_k}^{2e_k+d_k}.$$

Also by the Fundamental Theorem of Arithemtic, prime factorization is unique, so ordering correctly, for all i, $2f_i = 2e_i + d_i$. $d_i \geq 0$, so $2f_i \geq 2e_i$ and $f_i \geq e_i$. Appealing to Remark 5.20 in the notes, we now have that $a \mid b$. □

Question 5:

(a) Let x be an even integer. Modulo 8, the possible values for x are $0, 2, 4, 6$. We compute the square of each: 0, 4, 16, and 36, respectively. Taking each modulo 8, we get 0, 4, 0, and 4, respectively. Thus, $x^2 \equiv 0$ or 4 mod 8 when x is even.

(b) Similarly, let x be an odd integer. Modulo 8, the possible values of x are 1, 3, 5, or 7. We compute the square of each: 1, 9, 25, and 49, respectively. Taking each modulo 8, we get 1 for all values of x^2 $(9 - 8 = 1, 25 - 24 = 1, 49 - 48 = 1)$. Thus, $x^2 \equiv 1 \mod 8$ when x is odd.

Question 6: (a) *Proof.* We prove both directions of the if and only if. Let $3 \mid (d_k + \dots d_1 + d_0)$. This means that $d_k + \cdots + d_1 + d_0 \equiv 0 \mod 3$. Any power of 10 is equivalent to 1 modulo 3, thus we can mutliply every term by any integer in the form 10^k and not change the value modulo 3. So, we have $10^k d_k + \cdots + 10d_1 + d_0 \equiv 0 \mod 3$. So $n \equiv 0 \mod 3$, hence $3 \mid n$. The argument is symmetric. □

(b) $1+2+7+1+5+3+6+9+7+7+3+7+5 = 63$. $3 \mid 63$, and thus $3 \mid 102,715,369,770,375$.

Question 7: (a) We have that $A = g^a \mod 37 = 83,521 \mod 37 = 12$, and that $B = g^b \mod 37 = 410,338,673 \mod 37 = 15$. We calculate that S from both perspectives: $S = A^b \mod 37 = 12^7 \mod 37 = 9 = B^a \mod 37 = 15^4 \mod 37$.

(b) To find S^{-1}, we solve the equation $9S^{-1} \equiv 1 \mod 37$. $\gcd(9, 37) = 1$ (via the Euclidean Algorithm), so by Fermat's Little Theorem, we have that $9^{37-1} \equiv 1 \mod 37$. Thus, we get

$$9 \cdot 9^{35} \equiv 1 \mod 37.$$

So, $9^{35} \mod 37$ is S^{-1}. We compute that $9^{35} \equiv 33 \mod 37$. Thus, $S^{-1} \equiv 33 \mod 37$.

(c) Assuming Carlos used a $g = 17$ (even though he might not be since he is not using $p = 37$ like Alice and Bob), I used a Python script to find a value of c that satisfies $5 \equiv 17^c \mod 31$. I computed $c = 20$.

(d) I used a Python script to perform the following calculation for each of the characters in the secret message: for each character at index i of the encrypted message M, we solve for k in $9 \cdot k \equiv M_i \mod 37$. This gives a decoded message of 7DN292TWUZ.